

## VITORIA, Brazil

WMO: 836490

Lat: 20.258S

Lon: 40.286W

Elev: 11

StdP: 14.69

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 61.8       | 63.5 | 55.1                          | 64.5 | 70.3 | 57.0 | 69.3 | 70.3 | 21.4                  | 76.4 | 19.9 | 77.7 | 3.0                   | 300  | 0.351 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 2             | 14.0                      | 93.4            | 77.9 | 91.7 | 77.4 | 90.4 | 77.1 | 80.9                | 87.3 | 80.0 | 86.3 | 79.2 | 85.5 | 14.6                    | 30   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 79.2                            | 151.5 | 83.5 | 78.7 | 148.8 | 83.1 | 77.4 | 142.7 | 82.1 | 44.6          | 87.4 | 43.6 | 86.6 | 42.9 | 85.6 | 86.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 23.2              | 20.9 | 18.8 | 57.3                       | 97.5 | 3.3                | 2.0 | 54.9                                               | 99.0 | 53.0       | 100.1 | 51.1       | 101.3 | 48.7       | 102.7 |
|                   |      |      | 55.9                       | 82.8 | 2.4                | 1.9 | 54.1                                               | 84.2 | 52.7       | 85.3  | 51.3       | 86.4  | 49.6       | 87.8  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 77.8   | 82.3  | 82.9  | 81.8  | 79.5  | 75.9  | 74.0  | 73.0  | 73.6  | 75.2  | 77.4  | 78.1  | 80.6  |
|                                                                        | DBStd       | 4.77   | 2.88  | 2.83  | 2.99  | 3.12  | 3.07  | 3.10  | 3.12  | 3.40  | 3.72  | 4.06  | 4.16  | 3.49  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD50       | 10157  | 1000  | 921   | 986   | 885   | 802   | 720   | 714   | 731   | 756   | 848   | 844   | 950   |
|                                                                        | CDD65       | 4682   | 535   | 501   | 521   | 435   | 337   | 270   | 249   | 266   | 306   | 383   | 394   | 485   |
|                                                                        | CDH74       | 38005  | 5516  | 5420  | 5247  | 3851  | 2201  | 1511  | 1226  | 1356  | 1692  | 2636  | 2936  | 4413  |
|                                                                        | CDH80       | 11726  | 2022  | 2125  | 1862  | 1157  | 490   | 319   | 240   | 257   | 376   | 703   | 783   | 1392  |
| (14) Wind                                                              | WSAvg       | 8.3    | 10.0  | 8.7   | 7.7   | 6.8   | 6.5   | 6.3   | 6.8   | 7.9   | 9.7   | 10.0  | 9.7   | 9.8   |
| (15) Precipitation                                                     | PrecAvg     | 50.2   | 5.1   | 3.1   | 5.6   | 4.3   | 2.7   | 2.0   | 2.2   | 2.2   | 2.6   | 4.7   | 8.3   | 7.5   |
|                                                                        | PrecMax     | 77.7   | 11.8  | 9.4   | 16.8  | 15.6  | 9.7   | 6.9   | 5.0   | 8.3   | 6.4   | 17.5  | 25.6  | 27.0  |
|                                                                        | PrecMin     | 26.3   | 0.4   | 0.4   | 0.6   | 0.7   | 0.0   | 0.0   | 0.2   | 0.2   | 0.8   | 1.0   | 1.6   | 0.7   |
|                                                                        | PrecStd     | 13.1   | 3.2   | 2.1   | 3.9   | 3.0   | 2.2   | 1.6   | 1.3   | 1.7   | 1.7   | 3.5   | 5.4   | 5.0   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 94.6  | 95.3  | 94.8  | 93.3  | 91.0  | 89.3  | 88.0  | 87.9  | 89.9  | 91.7  | 93.2  | 93.8  |
|                                                                        |             | MCWB   | 79.0  | 78.8  | 78.5  | 77.5  | 75.1  | 74.0  | 72.0  | 72.5  | 73.3  | 75.2  | 76.3  | 78.2  |
|                                                                        | 2%          | DB     | 91.9  | 93.0  | 91.8  | 90.1  | 87.5  | 86.0  | 85.2  | 85.4  | 86.3  | 88.9  | 89.6  | 91.4  |
|                                                                        |             | MCWB   | 77.8  | 78.2  | 77.8  | 77.1  | 75.2  | 73.1  | 71.6  | 71.9  | 72.6  | 75.0  | 76.1  | 77.6  |
|                                                                        | 5%          | DB     | 90.4  | 91.2  | 90.1  | 88.0  | 84.6  | 83.8  | 82.6  | 82.8  | 84.2  | 86.3  | 87.5  | 89.4  |
|                                                                        |             | MCWB   | 77.3  | 77.8  | 77.7  | 77.0  | 74.3  | 72.8  | 71.4  | 71.5  | 72.3  | 74.5  | 76.1  | 77.3  |
|                                                                        | 10%         | DB     | 88.6  | 89.5  | 88.3  | 86.0  | 82.5  | 81.1  | 80.4  | 80.6  | 82.0  | 84.1  | 84.5  | 87.3  |
|                                                                        |             | MCWB   | 76.9  | 77.3  | 77.6  | 76.5  | 73.7  | 72.3  | 71.1  | 70.9  | 71.6  | 73.8  | 75.3  | 76.9  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 82.3  | 82.4  | 81.8  | 80.5  | 78.9  | 77.5  | 75.5  | 76.0  | 76.9  | 78.3  | 79.4  | 80.8  |
|                                                                        |             | MCDB   | 89.4  | 90.7  | 87.4  | 86.1  | 83.1  | 82.9  | 81.6  | 82.3  | 84.3  | 85.0  | 86.8  | 88.0  |
|                                                                        | 2%          | WB     | 80.2  | 80.8  | 80.5  | 79.4  | 77.5  | 75.6  | 74.2  | 74.5  | 75.3  | 77.0  | 78.2  | 79.5  |
|                                                                        |             | MCDB   | 86.8  | 87.3  | 86.5  | 85.3  | 82.7  | 81.2  | 79.8  | 80.5  | 80.8  | 83.7  | 85.4  | 86.3  |
|                                                                        | 5%          | WB     | 79.2  | 79.8  | 79.6  | 78.5  | 76.4  | 74.5  | 73.1  | 73.3  | 74.2  | 76.0  | 77.3  | 78.6  |
|                                                                        |             | MCDB   | 85.6  | 85.9  | 85.6  | 84.6  | 81.8  | 80.1  | 78.7  | 79.0  | 79.8  | 82.4  | 83.7  | 85.3  |
|                                                                        | 10%         | WB     | 78.4  | 79.0  | 78.8  | 77.6  | 75.3  | 73.4  | 72.1  | 72.3  | 73.1  | 75.1  | 76.4  | 77.8  |
|                                                                        |             | MCDB   | 84.8  | 85.0  | 84.8  | 83.5  | 80.5  | 79.1  | 78.0  | 78.1  | 78.9  | 81.3  | 82.1  | 84.1  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 13.2  | 14.0  | 13.5  | 13.1  | 13.7  | 14.7  | 14.9  | 14.5  | 13.2  | 12.5  | 11.4  | 12.3  |
|                                                                        |             | MCDBR  | 15.5  | 16.1  | 16.1  | 16.2  | 17.5  | 18.9  | 19.3  | 17.7  | 16.5  | 15.8  | 15.0  | 15.5  |
|                                                                        |             | MCWBR  | 5.7   | 5.7   | 6.0   | 6.5   | 8.0   | 8.8   | 8.7   | 7.7   | 6.7   | 6.2   | 5.9   | 6.1   |
|                                                                        | 5% WB       | MCDBR  | 13.7  | 14.3  | 14.1  | 13.8  | 15.3  | 16.1  | 16.6  | 14.9  | 13.9  | 13.7  | 13.2  | 13.4  |
|                                                                        |             | MCWBR  | 5.9   | 5.9   | 6.3   | 6.4   | 7.8   | 8.7   | 8.4   | 7.4   | 6.5   | 6.3   | 6.1   | 6.3   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.400 | 0.396 | 0.394 | 0.383 | 0.374 | 0.361 | 0.371 | 0.391 | 0.456 | 0.441 | 0.424 | 0.421 |
|                                                                        | taud        |        | 2.509 | 2.513 | 2.537 | 2.551 | 2.546 | 2.580 | 2.534 | 2.459 | 2.253 | 2.343 | 2.428 | 2.447 |
|                                                                        | Ebn at Noon |        | 300   | 298   | 290   | 281   | 271   | 269   | 270   | 274   | 267   | 281   | 291   | 294   |
|                                                                        | Edh at Noon |        | 36    | 36    | 34    | 31    | 29    | 27    | 29    | 33    | 44    | 42    | 39    | 39    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2020  | 2012  | 1764  | 1500  | 1244  | 1160  | 1212  | 1414  | 1521  | 1672  | 1651  | 1825  |
|                                                                        | RadStd      |        | 275   | 170   | 147   | 95    | 70    | 81    | 82    | 88    | 160   | 195   | 202   | 228   |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | -1.19   | N/A    | N/A     | -1.19 | -1.89 | N/A         | N/A   | N/A   | N/A   |
| Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page